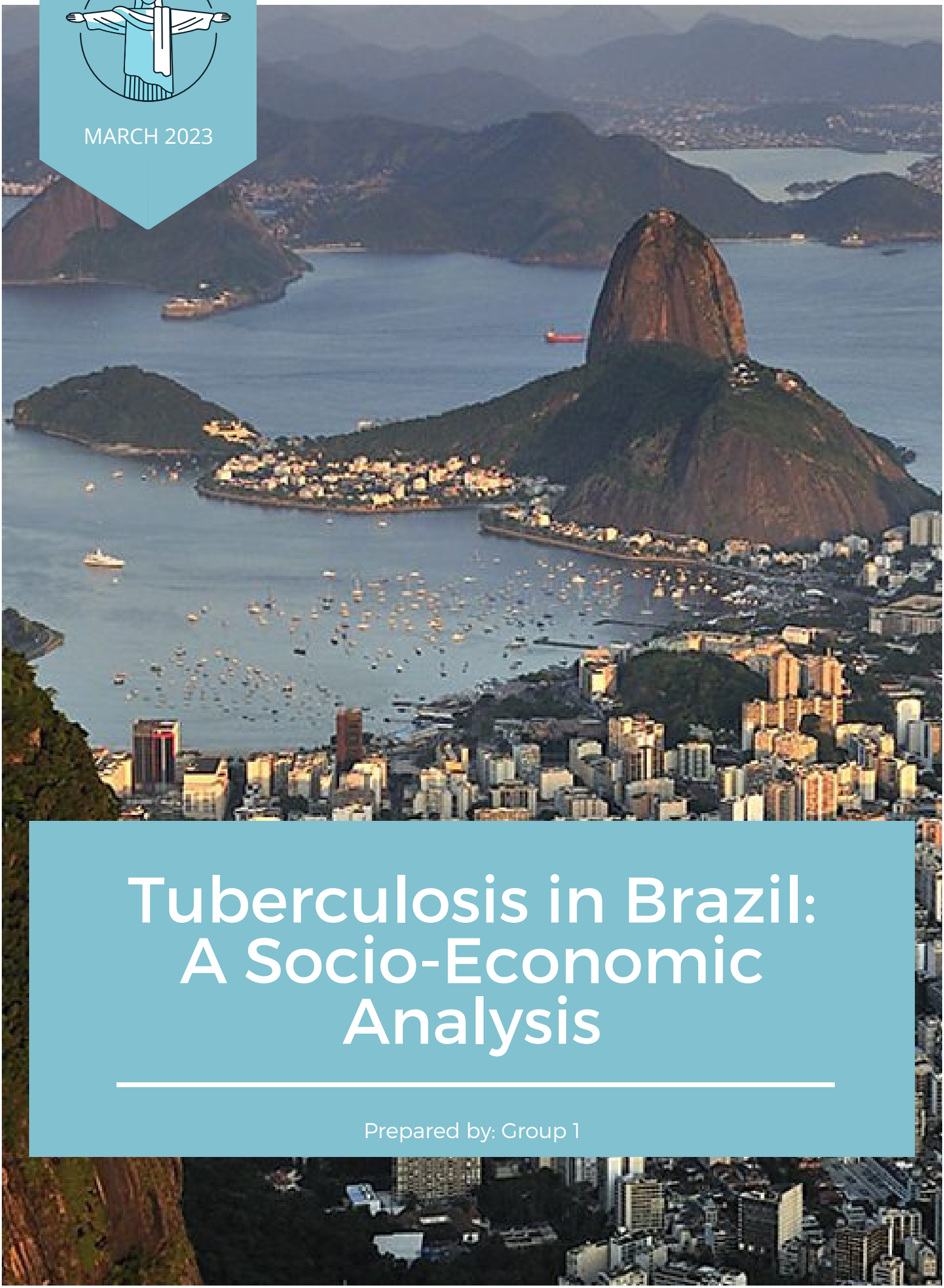


MARCH 2023

Tuberculosis in Brazil: A Socio-Economic Analysis

Prepared by: Group 1

EXECUTIVE SUMMARY

This report uses the provided data to examine the incidence of tuberculosis in Brazil. The study reveals that tuberculosis (TB) continues to be a serious public health concern in Brazil, with a high incidence rate and population prevalence rate. Also, the study finds that specific parts of Brazil, like the north and northeast, are disproportionately affected by tuberculosis.

Generalized Additive Models (GAMs) were used to investigate the relationship between tuberculosis (TB) incidence and socioeconomic covariates in Brazil, as well as to investigate the temporal and spatial distributions of TB risk in the country

The findings of the study indicate that tuberculosis in Brazil is strongly associated with Illiteracy, poverty, and poor living circumstances. In order to effectively combat TB in Brazil, the report suggests that the government invest in healthcare infrastructure and social programmes targeted at reducing poverty and enhancing living conditions.

Overall, the study emphasises the critical need for integrated government and healthcare provider efforts to address the TB epidemic in Brazil and reduce the disease burden on the community.

PRELIMINARY ANALYSIS

Cleaning and verifying data was the initial stage in data analysis. Checking for missing numbers, outliers, and data discrepancies. We used statistical approaches to eliminate or imputed missing or inconsistent data. We created a new variable TB Rate, which represents the number of tuberculosis cases per 100,000 people in each microregion and year.

Initially, we used a correlation matrix to determine the link between the variables in the dataset, and the following observations were made as a result.

1. There is a positive correlation between TB rate and Urbanisation, Density, Unemployment, and Timeliness.
2. There is a negative correlation between TB_rate and Illiteracy, Poverty, and Poor_Sanitation.
3. The correlation between TB_rate and the Indigenous population is weak.
4. Illiteracy, Poverty, and Poor_Sanitation are highly positively correlated with each other.
5. Urbanisation is negatively correlated with Illiteracy and poor sanitation but positively correlated with Density and Unemployment.
6. Density is positively correlated with the Indigenous population but negatively correlated with Illiteracy and Poverty.
7. Unemployment is negatively correlated with the Indigenous population but positively correlated with Urbanisation, Density, and Poverty.
8. Timeliness is negatively correlated with Illiteracy and poor sanitation but positively correlated with Urbanisation and Unemployment.

Preliminary investigations guided our Generalized Additive Model variable choices (GAM), and helped us choose which variables should be included in the GAM model by examining correlations and identifying the most significant predictors of TB incidence and prevalence.

MODEL SPECFICATION

To model Brazil's TB risk over time (2012-2014). After loading the data, we calculated the risk variable (TB/Population*10000).

We first expressed the covariates in a basic GLM. Due to its simplicity and nonlinear risk-covariable interactions, this model fit was poor. We tried GAMs. GAMs should assume a distribution. We plotted the risk distribution to find an acceptable distribution and found that the Poisson or normal distributions are more likely to help fit a reasonable model over the provided data.

The log-link function was used initially to model "y" as a Poisson distribution. Model fitting errors were considerable. Poisson model code execution was too long. The Poisson distribution expects discrete values, however, the response variable has continuous values. Hence, the Poisson (log) model was unsuitable for such data, explaining the poor fit. Because the computed response variable was non-negative, the additive model's normal distribution with an identity-link function was a good choice.

In our first normal distribution assumption, the GAM has eight smooth terms for each independent variable, We can then identify key phrases. The model's residuals were weak, but most factors were significant except illiteracy. The QQ plots and residual plots from all modelling attempts showed unsatisfactory residuals, thus we changed the type of "bs" and knots. To test linearity, we plotted risk against each variable. Unemployment, Poverty, Urbanisation, Density, and Timeliness were linear, while the others were non-linear. For spatial and temporal impacts, we included smooth functions for lon/lat and Year. The final model included an s(lon, lat,by=Year) smooth to show spatiotemporal issues. Fitting the final model:

$$Y_{[i]} \sim N(\mu, \sigma^2).Y_{[i]} \text{ indep}$$

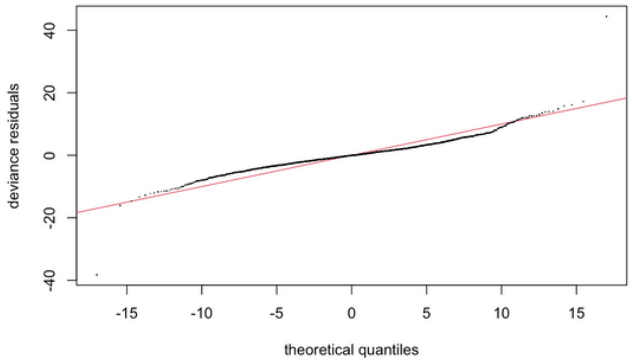
$$\mu_{[i]} = \beta_0 + \beta_1 \cdot x_1 + \beta_2 \cdot x_2 + \beta_3 \cdot x_3 + \beta_4 \cdot x_4 + \beta_5 \cdot x_5 + f_1(x_6) + f_2(x_7) + f_3(x_8) + f_4(x_9, x_{10})$$

where Y_i is the risk,x_1 is Unemployment, x_2 is Poverty, x_3 is Urbanization, x_4 is Density, x_5 is Timeliness, x_6 is Indigenous, x_7 is Illiteracy, x_8 is Poor_sanitation, x_9 is lon,x_{10} is lat and all f($\cdot$) are smooth functions defined by cubic regression splines.

ANALYSIS

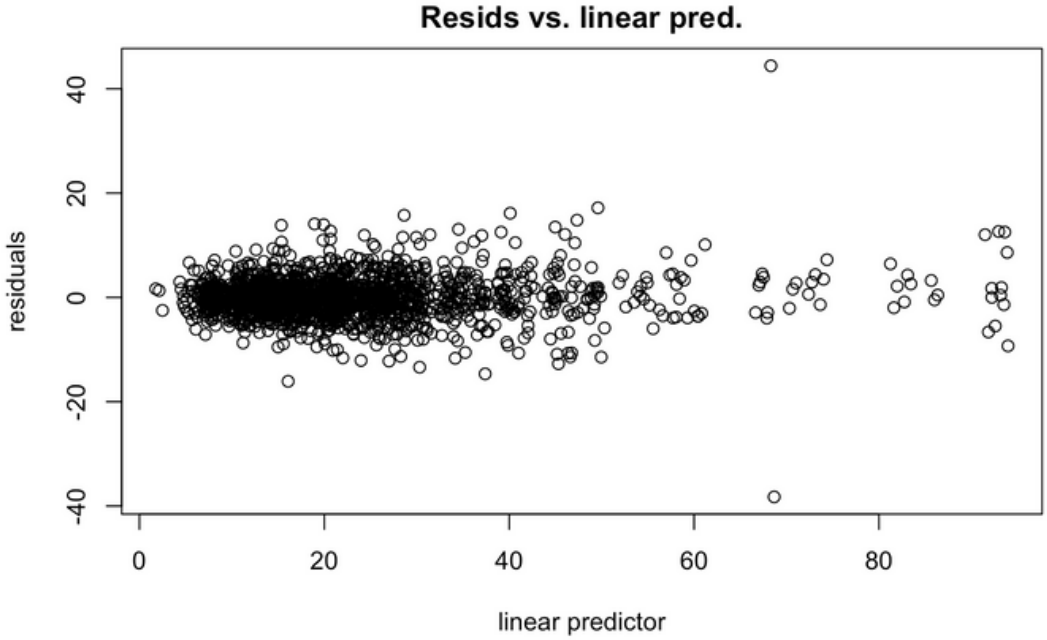
QQ PLOT

Since the data points closely resemble the diagonal line, the Q-Q plot reveals that the distribution of the sample data is roughly normal. However, the plot does deviate from the diagonal at a few points towards the right end. This suggests that some outliers or extreme values in the sample data may be responsible for deviations from normality.



RESIDUALS

With the exception of the tails, the model's residuals appear to closely match the theoretical normal distribution. It is possible that the model underestimates the distribution's extremes because the tails appear a bit thick. This might be as a result of the data having outliers. The residuals imply that the fitted values closely match the diagonal line on the top panel, demonstrating a strong fit between the model and the data. The fact that the dots are evenly distributed around the diagonal line suggests that the model captures the variation in the data adequately. The plot's extremes do contain a few anomalies, though. When examining the bottom panel, it is evident that residuals are frequently distributed randomly about the horizontal line at 0, indicating that the homoscedasticity and normality of residuals assumptions are largely satisfied. The regression plot indicates that the GAM model fits the data reasonably well.



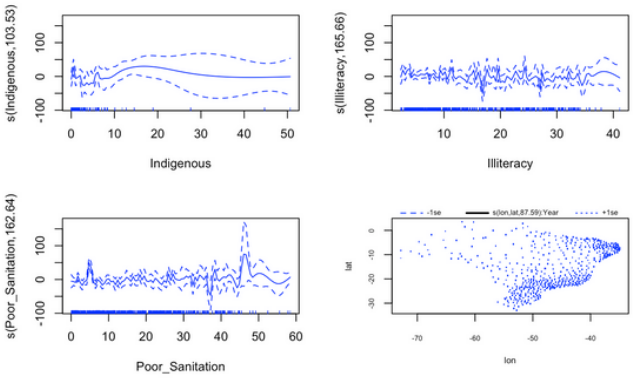
MODEL COMPARISON

Given that we have assessed the final model and are satisfied with the residual plots, our next step is to compare models in order to determine whether the final model is the best one thus far. To this end, we have chosen to use ANOVA for model comparison by testing the following hypotheses: H0 : the Former model is as good as the Final model vs. H1 : the Former model is NOT as good as the Final model the outcome indicates that the final model is significant better than the formar model with the p-value< 2.2e-16 ***.

ANALYSIS CONT..

SMOOTH TRENDS PLOT

Looking at the smooth terms plot, we can see that the covariate “Indigenous” has a nonlinear relationship with TB_rate, with a peak at around 0.4 on the x-axis. “Illiteracy” also has a nonlinear relationship, with a peak at around 0.2 on the x-axis. “Poor_Sanitation” shows a gradual increase in TB_rate as the covariate value increases. The smooth effect for the interaction between “lon” and “lat” with “Year” appears to show a complex spatio-temporal pattern in TB_rate. The remaining covariates (“Unemployment”, “Poverty”, “Urbanisation”, “Density”, and “Timeliness”) all appear to have linear relationships with TB_rate.



After analysing the aforementioned variables, we decided to plot maps in order to investigate the predicted TB rates in Brazil. This resulted in the below map, which displays TB rates by region. The map is used to represent the spatial distribution of a variable over a geographic region by dividing the region into small areas (such as states, provinces, or counties). In this instance, the map depicts the distribution of tuberculosis (TB) rates across the states of Brazil.



RESULTS

GAM ANALYSIS

The GAM was used to determine the relationship between socio-economic covariates and the rate of TB per unit population. The model summary and plots were examined to answer the following questions:

HOW WELL DOES THE MODEL FIT THE DATA?

The residual plot showed how well the model suited the data variability. The residual plot displays an even residual distribution, indicating the model fits the data well. The model may not capture all data variability since certain residuals diverge from the expected range. The model's R-squared value was 0.708, showing that socio-economic factors explain 70.8% of TB incidence rates.

ARE ANY OF THE SOCIO-ECONOMIC COVARIATES SIGNIFICANTLY AFFECTING THE RATE OF TB PER UNIT POPULATION?

The results indicate that some socioeconomic covariates have a significant impact on the rate of TB per unit of the population. According to the parametric coefficients in the model summary, Unemployment, Urbanization, and Timeliness are significantly associated with TB rates, with p-values less than 0.05. Additionally, the smooth terms plot shows that Indigenous, Illiteracy, Poor Sanitation, and the spatiotemporal effect (long, lat:Year) have a significant impact on TB incidence rates, with curves deviating from zero and p-values of 0.001.

IN WHAT WAY ARE THE SOCIO-ECONOMIC COVARIATES AFFECTING THE RATE OF TB PER UNIT POPULATION?

The smooth terms plot indicates that socioeconomic covariates influence the incidence of tuberculosis per unit of population in varying ways. Indigenous, Illiteracy and Poor Sanitation have a positive correlation with the TB rate, suggesting that as these covariates increase, so does the TB rate. In contrast, Unemployment and Urbanization have a negative correlation with the TB rate, which means that as these covariates increase, the TB rate tends to decrease. Notably, the relationships may not be causal, and other factors not accounted for in the model may be influencing the TB rate.

WHAT IS THE SPATIAL STRUCTURE OF ANY SYSTEMATIC (STRUCTURED) RISK THAT IS NOT EXPLAINED BY THE COVARIATES?

In terms of the spatial structure of this risk, it appears that the predicted rate of tuberculosis cases is concentrated in certain regions, as opposed to being randomly distributed across the country This suggests that there may be spatially-related factors contributing to the higher TB rates in these areas that are not captured by the model's covariates.

ARE THERE REGIONS WHERE THE RATE OF TB PER UNIT POPULATION IS HIGH AND WHERE MORE RESOURCES SHOULD BE ALLOCATED?

We recommend dedicating resources to Brazil's high-TB regions. Northern and northeastern Brazil have higher TB rates. The maps show regions with high anticipated TB rates. These regions contain Amazonas, Roraima, Amapá, Pará, Maranho, and Bahia. These areas are ideal for tuberculosis resource allocation. These hospitals should receive assistance from the health authorities to handle the increasing tuberculosis patients.

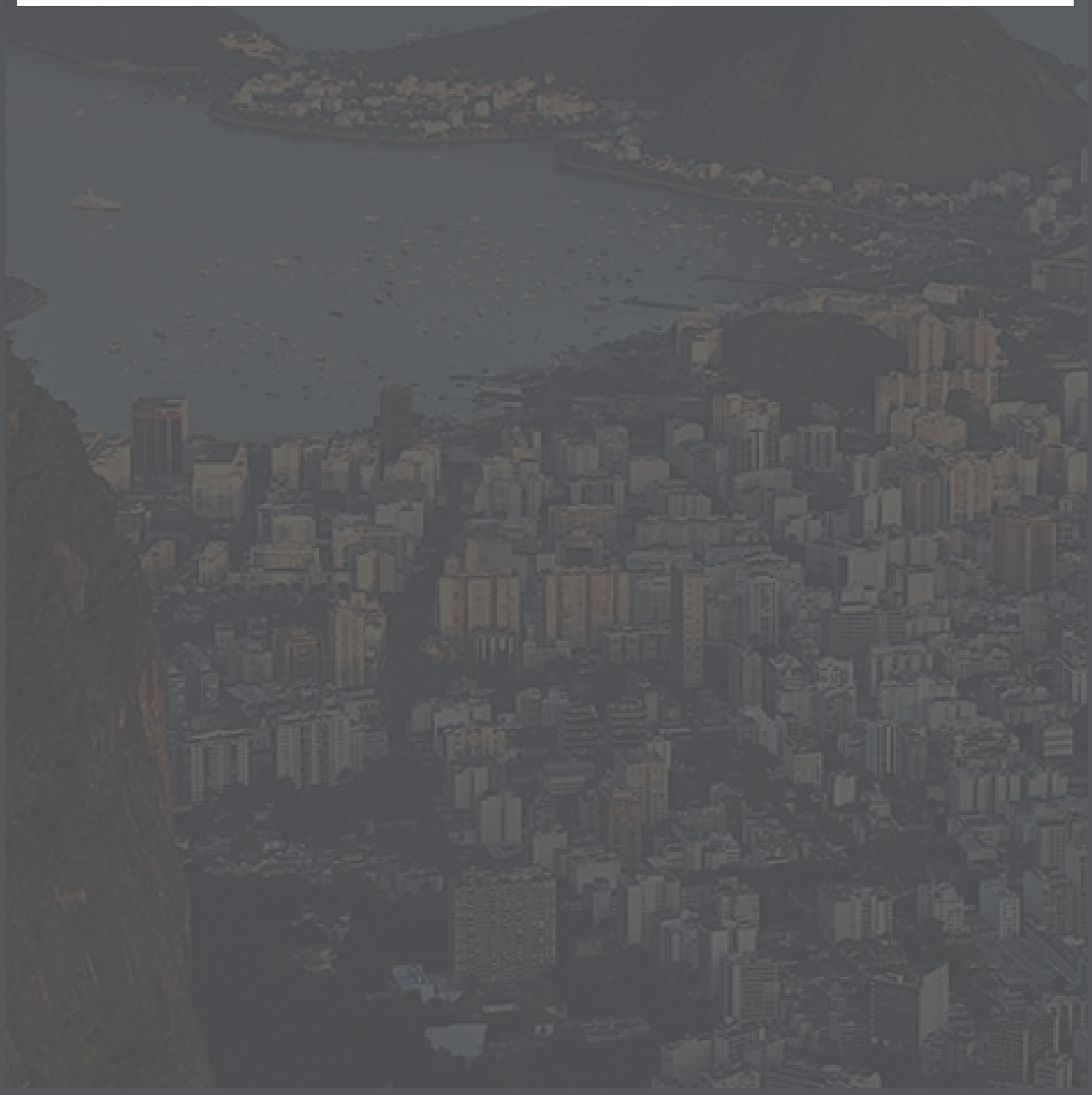
WHAT ARE THE LIMITATIONS OF THE STUDY?

The use of cross-sectional data limits the ability to establish causality, which is one of the study's limitations. In addition, the model does not account for all socioeconomic covariates that may influence tuberculosis (TB) incidence rates, and there may be additional variables not accounted for. Lastly, the study focuses solely on the association between socioeconomic covariates and TB incidence rates, whereas environmental and genetic factors may also influence TB incidence rates.

After analyzing the results of the model and residuals, it was concluded that changing the link function to log would give an optimal model fit. By adding the log link to the gaussian distribution, it constricts the values for the response variable to be non-negative which better respects the nature of the response, leading to a better model

CONCLUSION

This study found that socioeconomic factors, such as unemployment, urbanisation, and poor sanitation, are significant predictors of TB incidence rates in Brazil. Additionally, the study revealed that Indigenous, Illiteracy, Poor Sanitation, and the spatiotemporal effect (lon,lat:Year) have a significant influence on TB incidence rates. These findings have significant ramifications for public health policy and practice and highlight the need to consider socioeconomic factors when designing interventions to prevent and treat tuberculosis. The complex relationship between socioeconomic factors and incidence rates of tuberculosis in Brazil requires additional study



CODE

```
#Loading required libraries

library(sf)
library(tidyverse)
library(ggplot2)
library(dplyr)
library(mgcv)
library(sp)
library(fields)
library(maps)

#Importing Dataset
Data = load("/Users/minahilashfaq/Desktop/506-Group/datasets_project.RData")
head(Data)

# calculate outliers for unemployment variable
Unemployment_outliers <- boxplot.stats(TBdata$Unemployment)$out

# update the dataframe without outliers
TBdata <- TBdata[!TBdata$Unemployment %in% Unemployment_outliers,]

# create boxplot of the new dataframe to check
boxplot(TBdata$Unemployment)

# Calculate the TB rate (cases per unit population)
TBdata$TB_rate <- TBdata$TB / TBdata$Population * 100000

# View the first few rows of the dataframe to confirm the new variable has been added
head(TBdata)

# Create a new data frame with only the relevant variables
TBdata_subset <- TBdata[, c("TB_rate", "Indigenous", "Illiteracy", "Urbanisation", "Density", "Poverty", "Poor_Sanitation", "Unemployment", "Timeliness")]

# Calculate the correlation matrix
cor_matrix <- cor(TBdata_subset)

# Print the correlation matrix
print(cor_matrix)
```

	TB_rate	Indigenous	Illiteracy	Urbanisation	Density	Poverty	Poor_Sanitation	Unemployment
TB_rate	1.00000000	0.12825598	-0.18145110	0.29477284	0.19464746	-0.09061062	-0.2126075	0.31701168
Indigenous	0.12825598	1.00000000	0.07085347	-0.17272080	0.45204314	0.09509612	0.2381742	0.04760431
Illiteracy	-0.18145110	0.07085347	1.00000000	-0.70996024	0.40094056	0.89762937	0.7069094	0.35397475
Urbanisation	0.29477284	-0.17272080	-0.70996024	1.00000000	-0.42974472	-0.75711389	-0.8345401	-0.02079022
Density	0.19464746	0.45204314	0.40094056	-0.42974472	1.00000000	0.58612416	0.5929751	0.38571182
Poverty	-0.09061062	0.09509612	0.89762937	-0.75711389	0.58612416	1.00000000	0.7739168	0.50441698
Poor_Sanitation	-0.21260751	0.23817416	0.70690943	-0.83454012	0.59297510	0.77391676	1.0000000	0.10801565
Unemployment	0.31701168	0.04760431	0.35397475	-0.02079022	0.38571182	0.50441698	0.1080157	1.00000000
Timeliness	0.30864372	-0.02384079	-0.56810092	0.53028899	-0.08234903	-0.50402108	-0.4045976	-0.05485677

	Timeliness
TB_rate	0.30864372
Indigenous	-0.02384079
Illiteracy	-0.56810092
Urbanisation	0.53028899
Density	-0.08234903
Poverty	-0.50402108
Poor_Sanitation	-0.40459762
Unemployment	-0.05485677
Timeliness	1.00000000

```
#Fitting the initial model
formula0 <- TB_rate ~ s(Indigenous,bs="cs")+ s(Unemployment,bs="cs")+
s(Illiteracy,bs="cs")+ s(Poverty,bs="cs")+ s(Urbanisation,bs="cs")+ s(Density,bs="cs")+ s(Poor_Sanitation,bs="cs")+ s(Timeliness,bs="cs")

model0 <- gam(formula, data = TBdata, family = gaussian(link = 'log'), scale = -1, gamma = 1.4)

#Checking the model summary
summary(model0)

#Plotting the model
gam.check(model0)
plot(model0)
```

```r
#Fitting the final model
formula <- TB_rate ~ s(Indigenous,k=250,bs="cr")+ s(Illiteracy,k=300,bs="cr") +
            Unemployment + Poverty + Urbanisation + Density + Timeliness +
            s(Poor_Sanitation, k=250, bs="cr") +
            s(lon, lat, by=Year, k=300)

model <- gam(formula, data = TBdata, family = gaussian(link = 'identity'), scale = -1, gamma = 1.4)

#Checking the model summary
summary(model)

#Plotting the model
gam.check(model)
plot(model)

#Comparing the initial and final model
anova(model0,model)
```

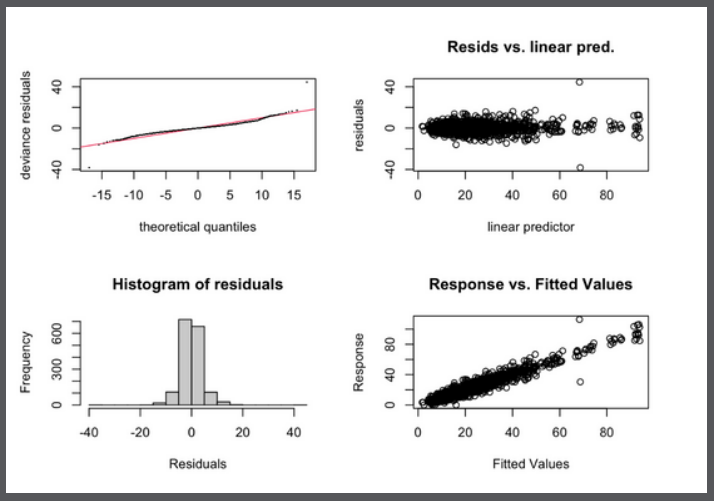
```
Family: gaussian
Link function: identity

Formula:
TB_rate ~ s(Indigenous, k = 250, bs = "cr") + s(Illiteracy, k = 300,
    bs = "cr") + Unemployment + Poverty + Urbanisation + Density +
    Timeliness + s(Poor_Sanitation, k = 250, bs = "cr") + s(lon,
    lat, by = Year, k = 300)

Parametric coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  692.64950  302.45826   2.290   0.0222 *
Unemployment   2.61381    0.52009   5.026 5.84e-07 ***
Poverty        0.21513    0.15118   1.423   0.1550
Urbanisation   0.45618    0.10967   4.160 3.43e-05 ***
Density        7.39388   14.55697   0.508   0.6116
Timeliness     0.07233    0.04200   1.722   0.0853 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Approximate significance of smooth terms:
                     edf Ref.df     F  p-value
s(Indigenous)      103.53 105.71 2.549  < 2e-16 ***
s(Illiteracy)      165.66 170.44 1.883  < 2e-16 ***
s(Poor_Sanitation) 162.64 167.10 2.482  < 2e-16 ***
s(lon,lat):Year     87.59  93.79 1.812 8.56e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

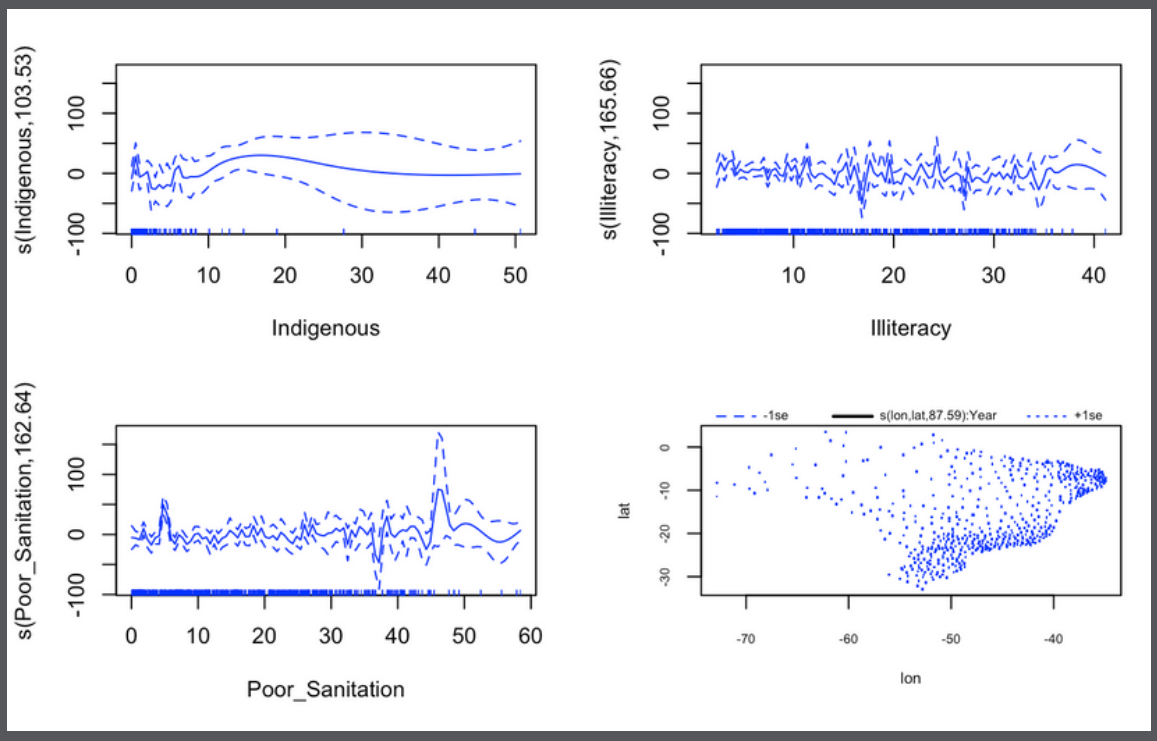
R-sq.(adj) =   0.89   Deviance explained = 92.5%
GCV = 55.073  Scale est. = 24.61     n = 1638
```




```
Analysis of Deviance Table

Model 1: TB_rate ~ s(Indigenous, k = 100, bs = "cr") + s(Illiteracy, k = 100,
    bs = "cr") + Unemployment + Poverty + Urbanisation + Density +
    Timeliness + s(Poor_Sanitation, k = 100, bs = "cr") + s(lon,
    lat, by = Year, k = 80)
Model 2: TB_rate ~ s(Indigenous, k = 250, bs = "cr") + s(Illiteracy, k = 300,
    bs = "cr") + Unemployment + Poverty + Urbanisation + Density +
    Timeliness + s(Poor_Sanitation, k = 250, bs = "cr") + s(lon,
    lat, by = Year, k = 300)
  Resid. Df Resid. Dev     Df Deviance
1    1304.6      62486
2    1095.0      27380 209.61    35106
```

```r
# Plot smooth terms
plot(model, pages = 1, col = "blue", se = TRUE)
```



```
# Convert TBdata to spatial points dataframe
TB_sp <- st_as_sf(TBdata, coords = c("lon", "lat"), crs = 4326)
TB_sp <- st_transform(TB_sp, 4326)

# Extract lon and lat values
TB_sp$lon <- st_coordinates(TB_sp)[,1]
TB_sp$lat <- st_coordinates(TB_sp)[,2]

# Make a prediction for each location
TB_sp$pred_rate <- predict(model, newdata = TB_sp, type = "response")

# Extract the predicted rates and convert to a dataframe
pred_df <- data.frame(TB_sp$pred_rate)
colnames(pred_df) <- "pred_rate"
TB_sp_df <- cbind(as.data.frame(TB_sp), pred_df)

# Sort by longitude and latitude
TB_sp_df <- TB_sp_df[order(TB_sp_df$lon, TB_sp_df$lat),]

# Filter out missing values
TB_sp_df <- na.omit(TB_sp_df)

# Check if the coordinates are within the map boundaries
TB_sp_df <- TB_sp_df[TB_sp_df$lon > -75 & TB_sp_df$lon < -30 & TB_sp_df$lat > -35 & TB_sp_df$lat < 10, ]

# Define color ramp
my_colors <- colorRampPalette(c("#ffffcc", "#a1dab4", "#41b6c4", "#2c7fb8", "#253494","#FF0000" ))

# Set up the plotting region
plot.window(xlim = c(-80, -25), ylim = c(-40, 15))
par(mar = c(2, 2, 1, 3), xaxs = "i", yaxs = "i")

# Plot the map and points
filled.contour(x = sort(unique(TB_sp_df$lon)), y = sort(unique(TB_sp_df$lat)),
          z = matrix(TB_sp_df$pred_rate, nrow = length(unique(TB_sp_df$lat)), ncol = length(unique(TB_sp_df$lon))),
          color.palette = my_colors, xlab = "Longitude", ylab = "Latitude",
          main = "\nPredicted TB Rate\n", xlim = c(-75, -30), ylim = c(-35, 10), axes = FALSE)
axis(1, at = seq(-75, -30, by = 5))
axis(2, at = seq(-35, 10, by = 5))

# Add the map boundary
maps::map("world", "brazil", xlim=c(-75,-30), ylim=c(-35,10), add = TRUE, fill = FALSE, lty = 1, col = "gray50", lwd = 2, mar = c(0, 0, 0, 0))

#Adding points on the map
points(TB_sp_df$lon, TB_sp_df$lat, pch = 25, bg = my_colors(10)[cut(TB_sp_df$pred_rate, breaks = 10)],
     cex = sqrt(TB_sp_df$pred_rate)/max(sqrt(TB_sp_df$pred_rate)), lwd = 0.5)

# add legend outside of plot area
legend("topright", legend = round(seq(0, max(TB_sp_df$pred_rate), length.out = 11), 1), fill = my_colors(10), title = "TB Rate", cex = 0.8, box.lty = 0, horiz = TRUE,
x.intersp = 0.7, y.intersp = 0.7)
```

